

Introduction to Human Anatomy

Anatomical terms L1

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ILOs

- **Describe the anatomical position and the anatomical planes**
- **Define the anatomical terms of position and movements**

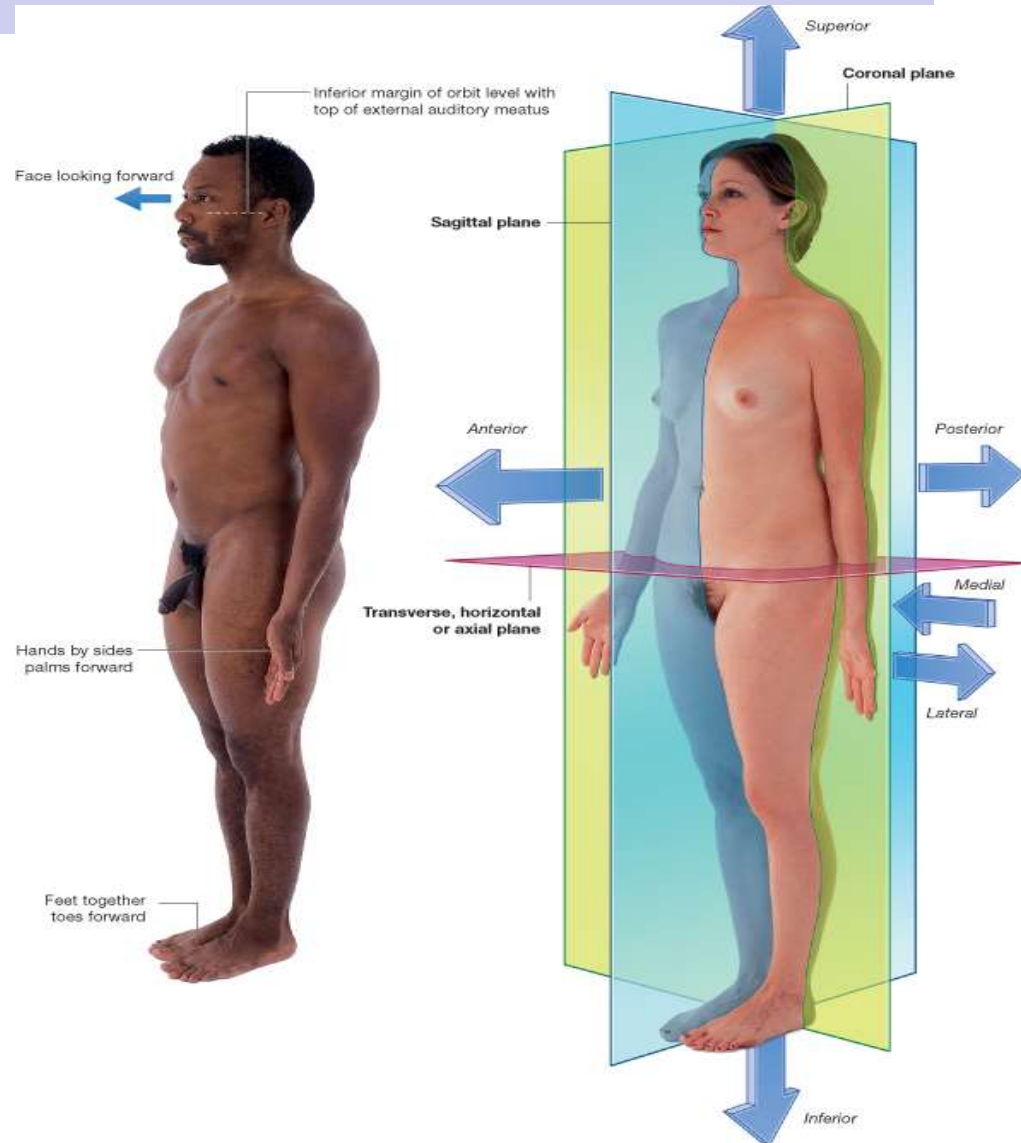
Anatomical Terminology

Anatomy:

Study of morphological structure of the body

Anatomical position

- Body is standing erect
- Face looking forward
- Upper limb hanging by side
- hands facing forwards
- Lower limbs are parallel



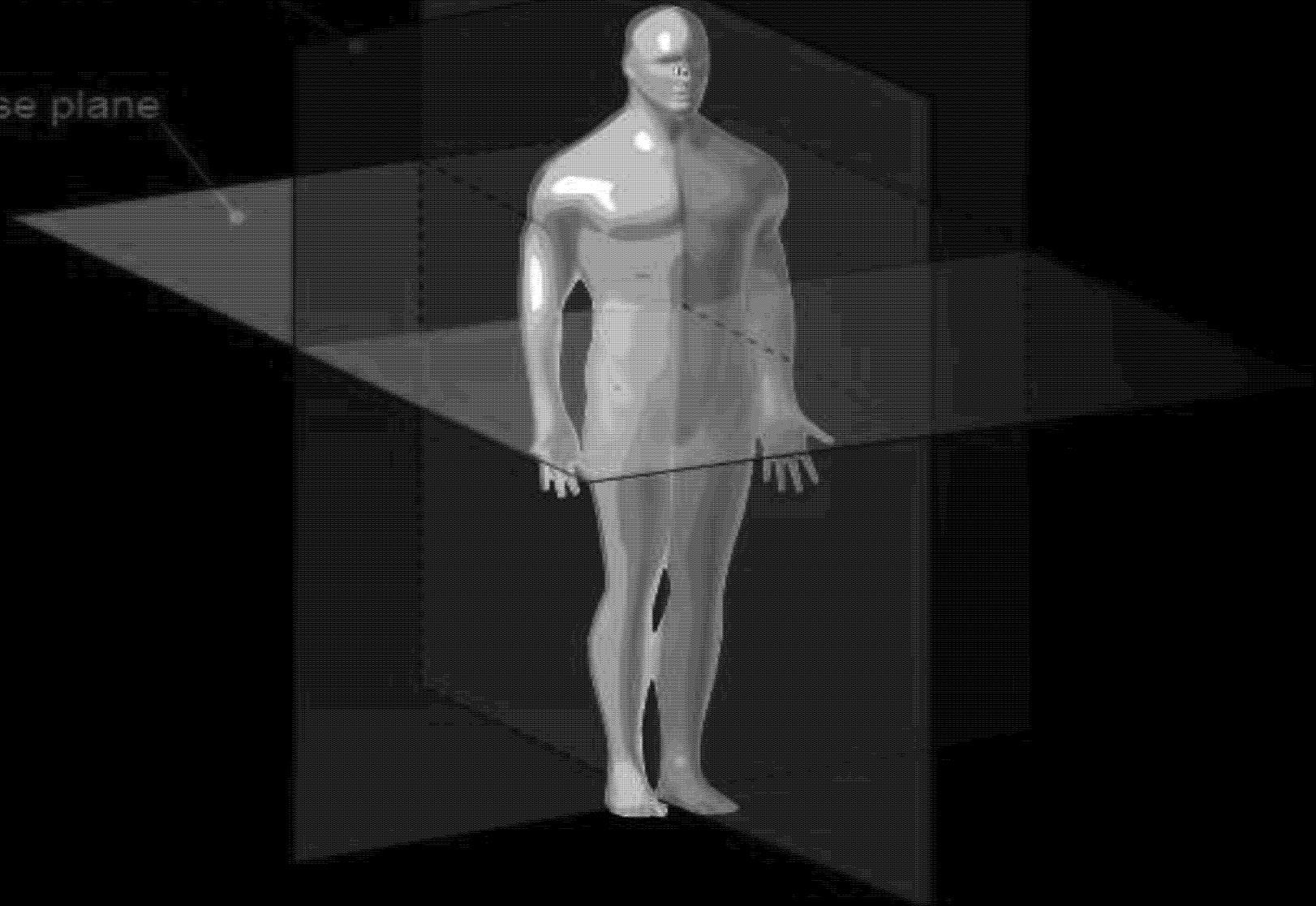
Anatomical Planes

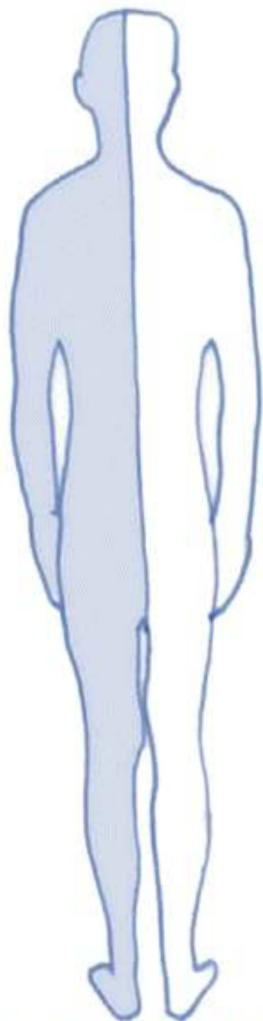
- **Median plane:** Vertical plane passing through center of body dividing it into **2 equal halves(right and left)**
- **Paramedian plane:** Vertical plane dividing body into **2 unequal halves**
- **Coronal plane:** Vertical plane perpendicular on median plane dividing body **into anterior& posterior**
- **Transverse:** Horizontal plane dividing body to **upper and lower parts**

Sagittal plane

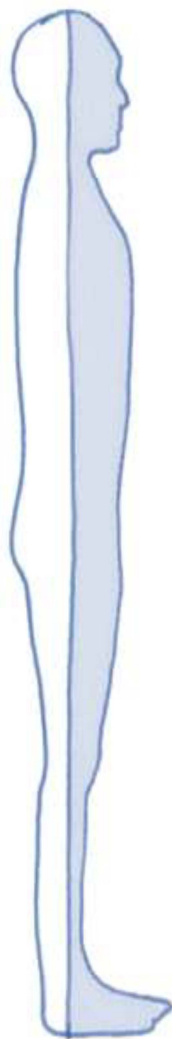
Coronal plane

Transverse plane

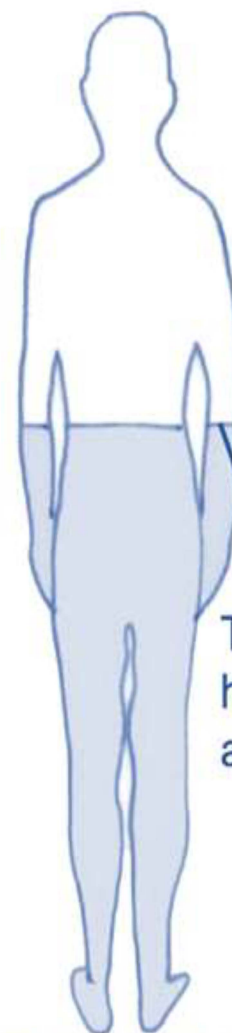




Median/Sagittal plane



Frontal/coronal plane



Transverse/
horizontal
axis

Transverse/horizontal
plane

Anatomical Terms

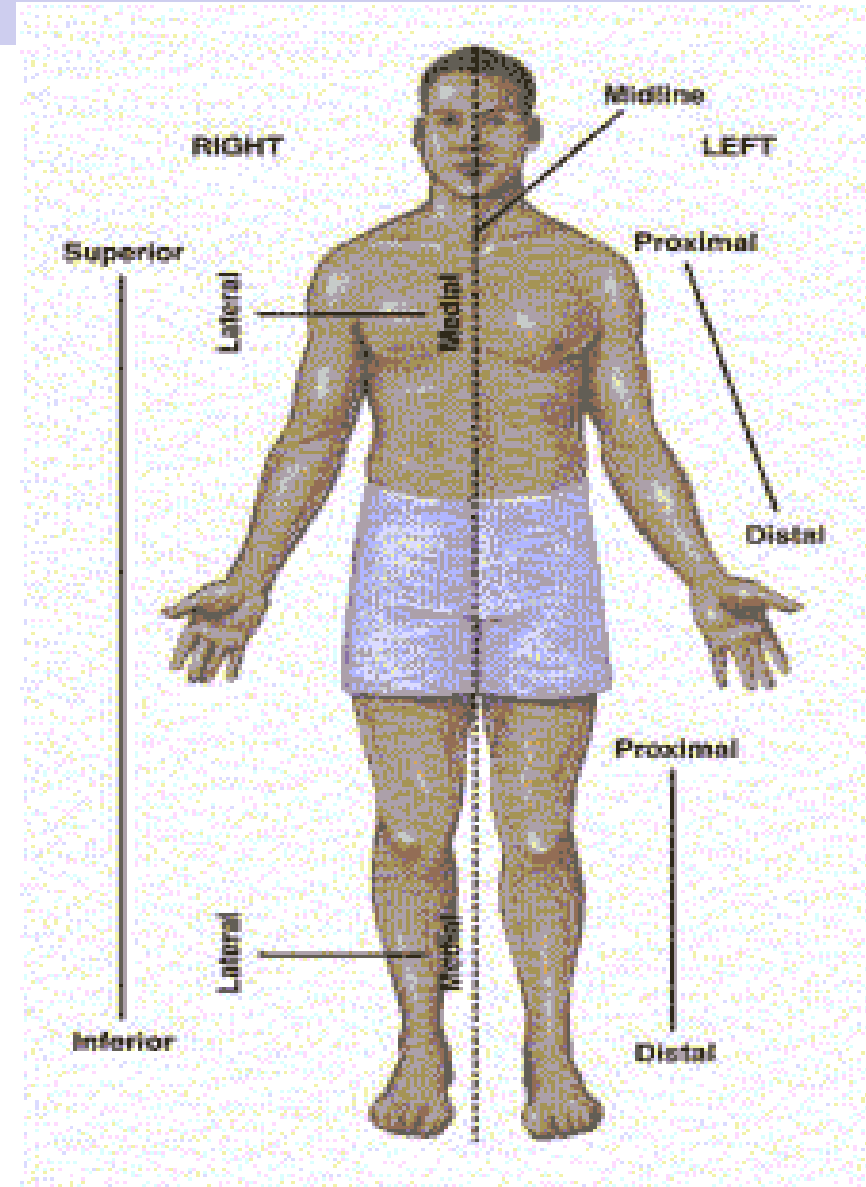
Median: structure lies **in** median plane

Medial : structure lies **nearer to** median plane

Lateral: structure lies **away from** median plane

Superior: close to upper end of body= skull

Inferior: close to lower end of body= floor



Anatomical Terms

- **Anterior:**

structure is nearer to **front**

- **Posterior:**

structure is nearer to **back**

- **Superficial :**

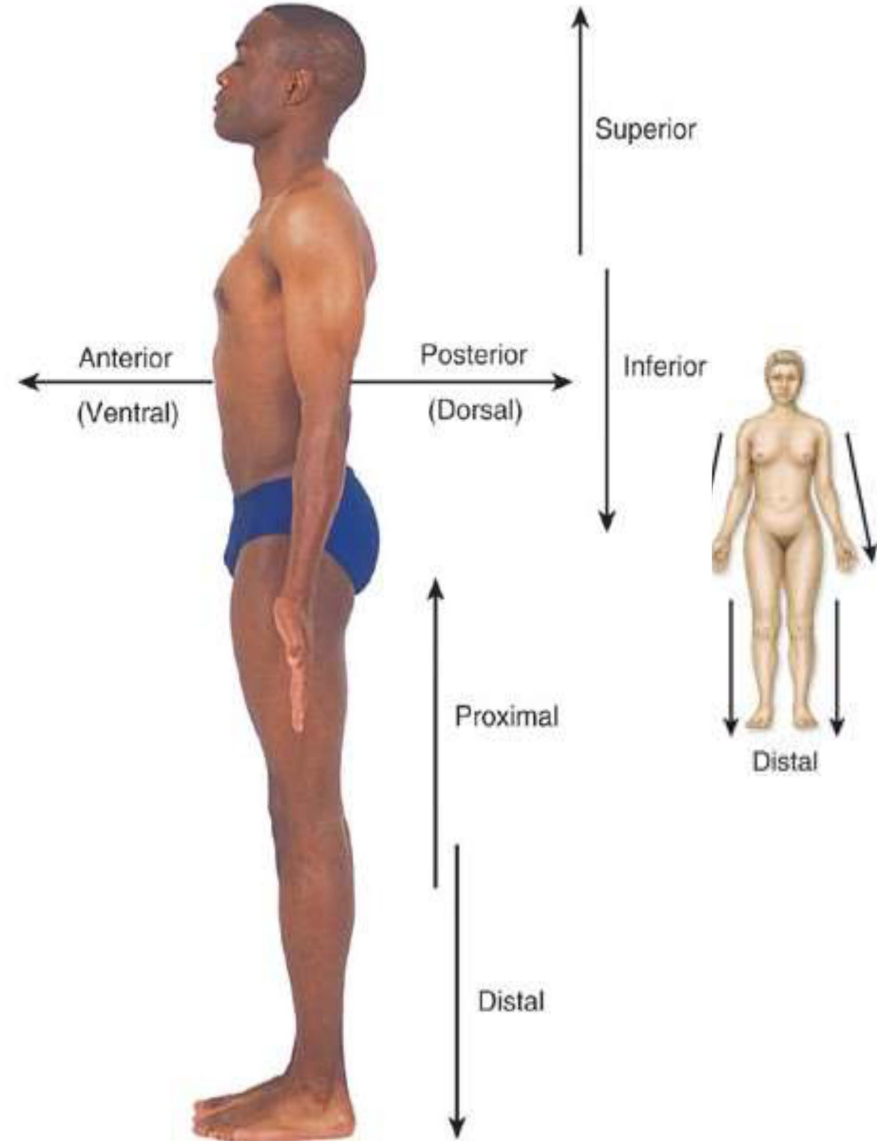
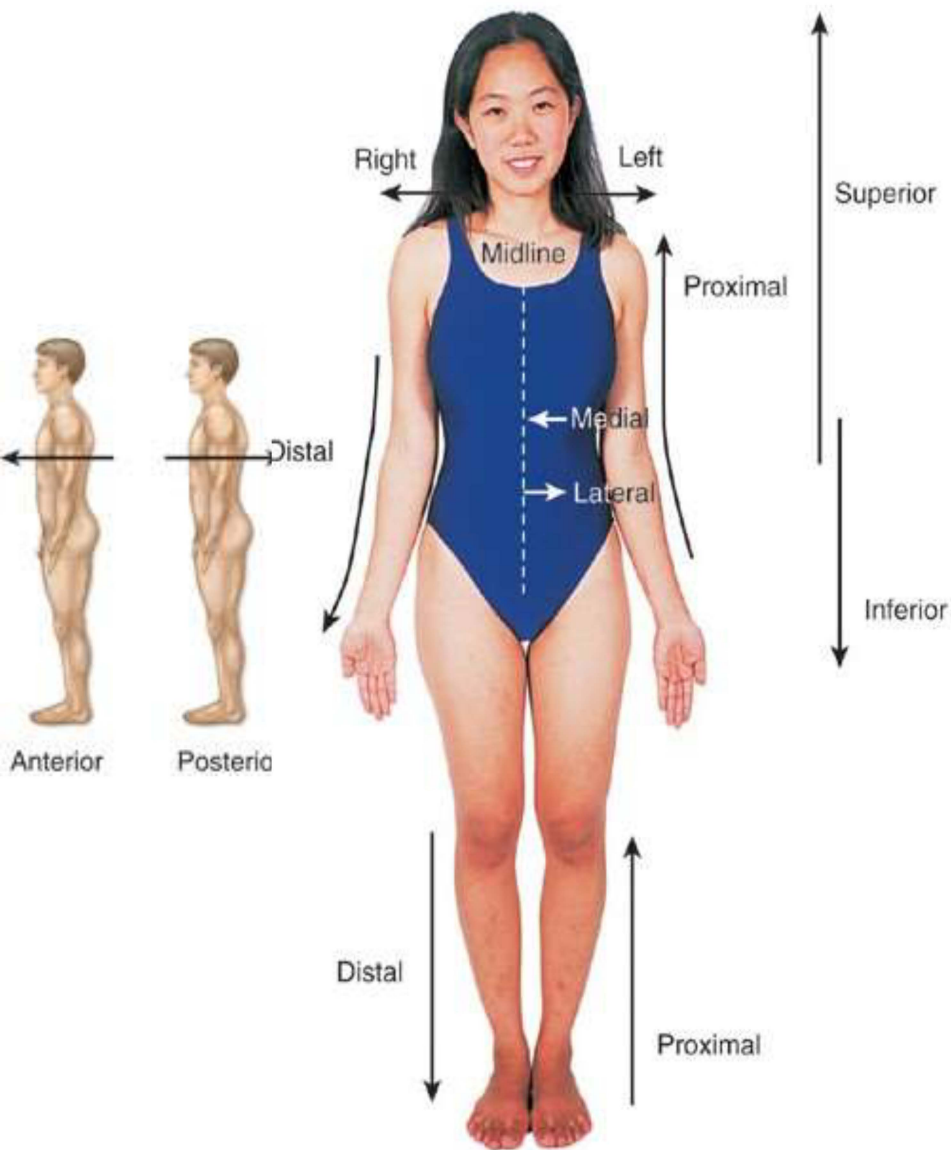
structure **nearer to surface**

- **Deep:**

structure **away from surface**

- **Internal :**

structure is **inside** organ or cavity



Proximal:

Structure is nearer to trunk

Distal :

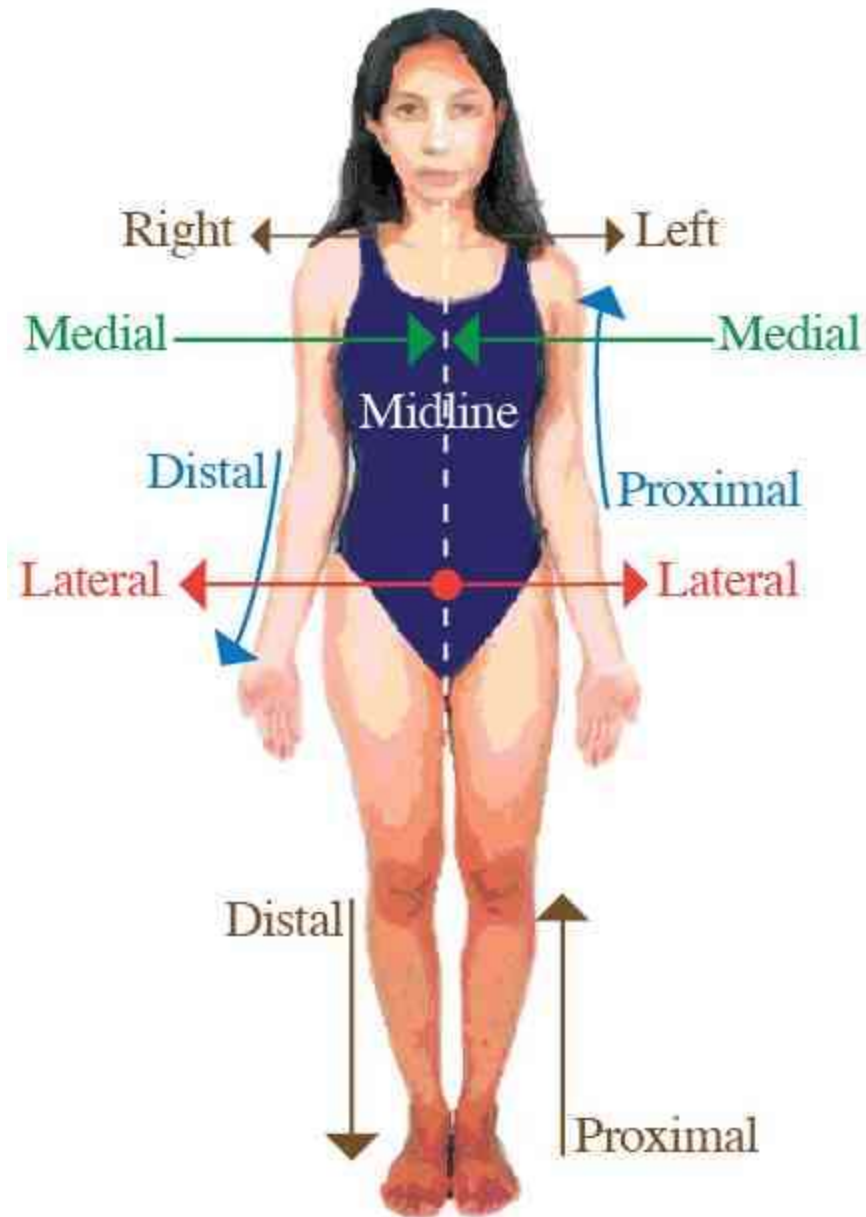
Structure is away from trunk

Ipsilateral:

Structure is on same side

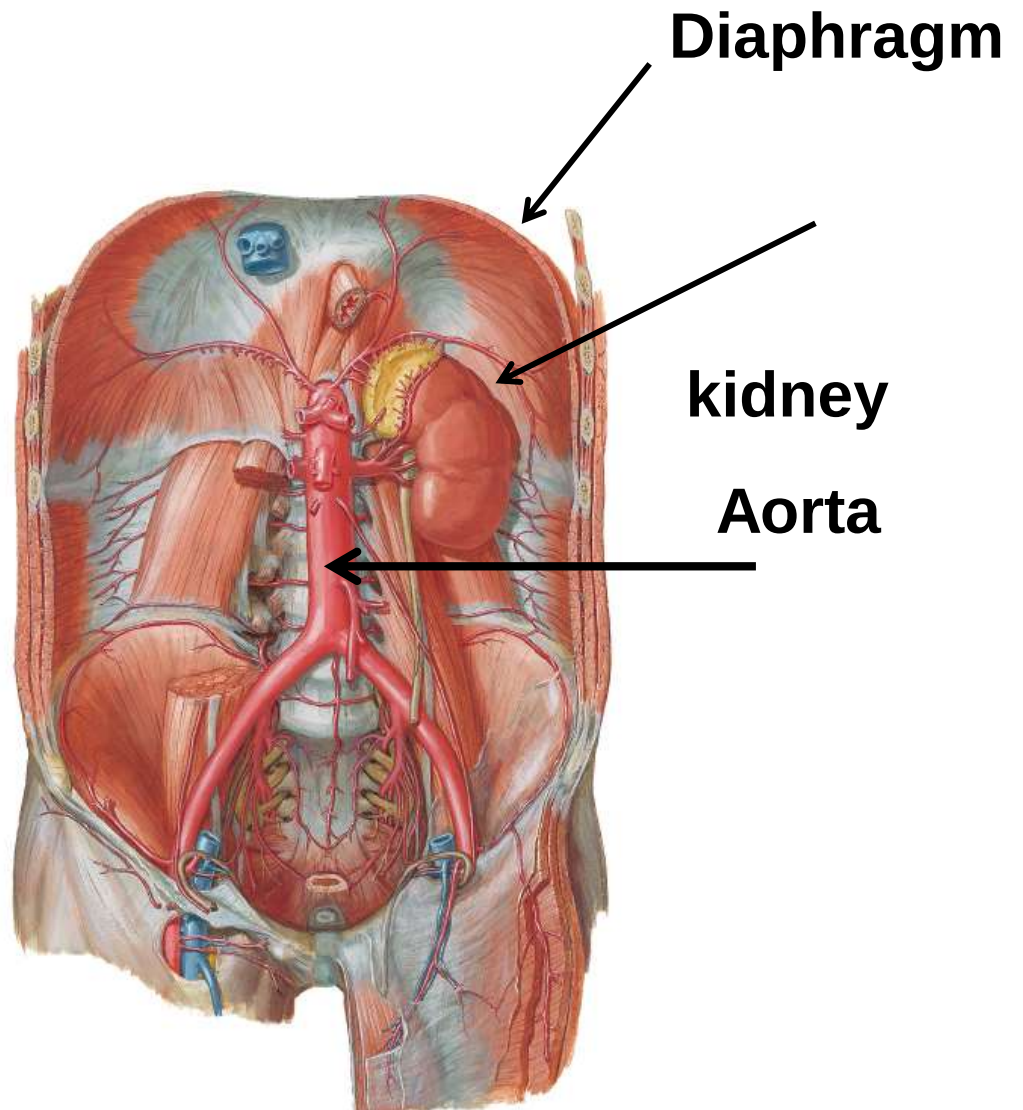
Contralateral:

Structure on opposite side

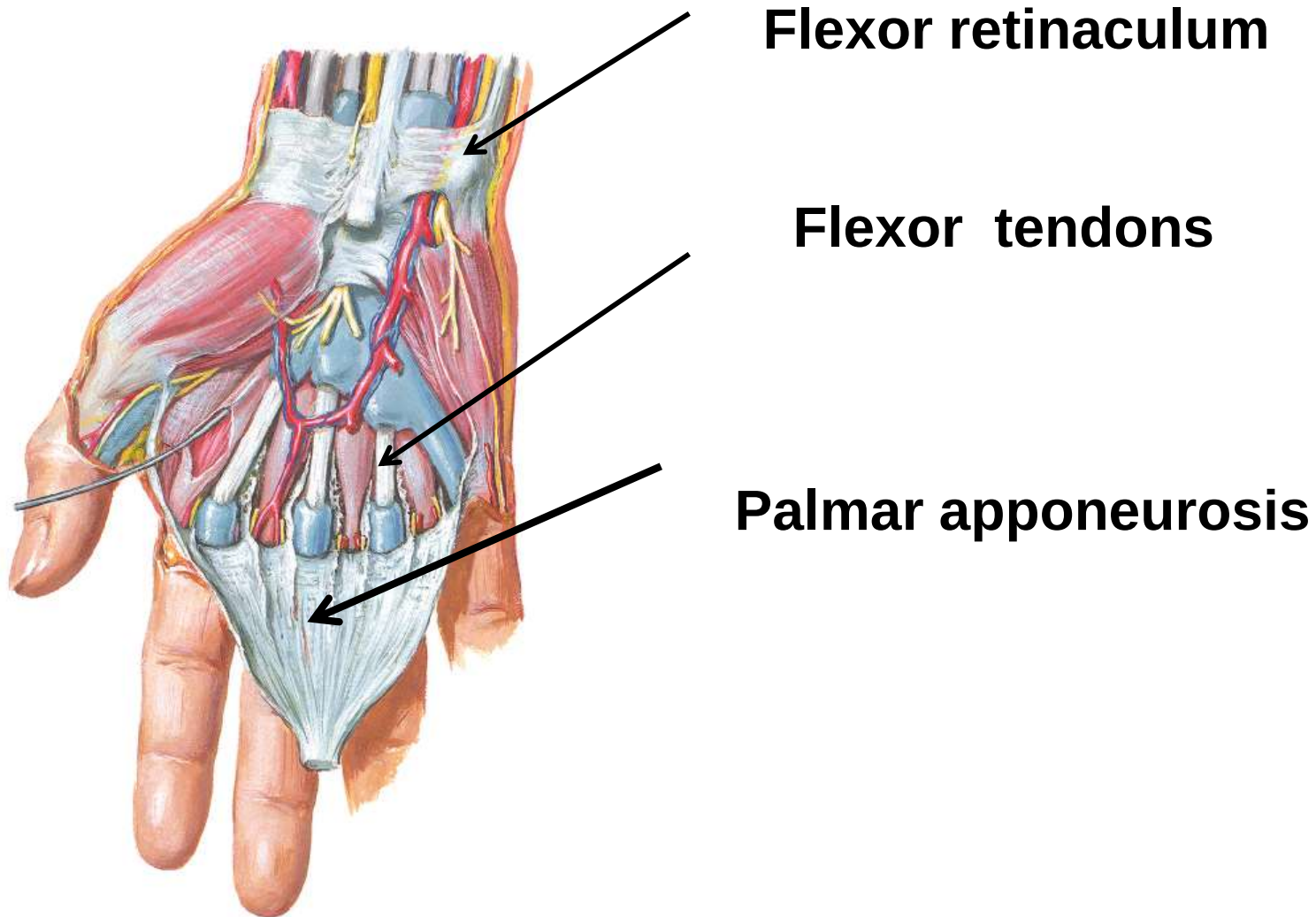


Interactive Exercise

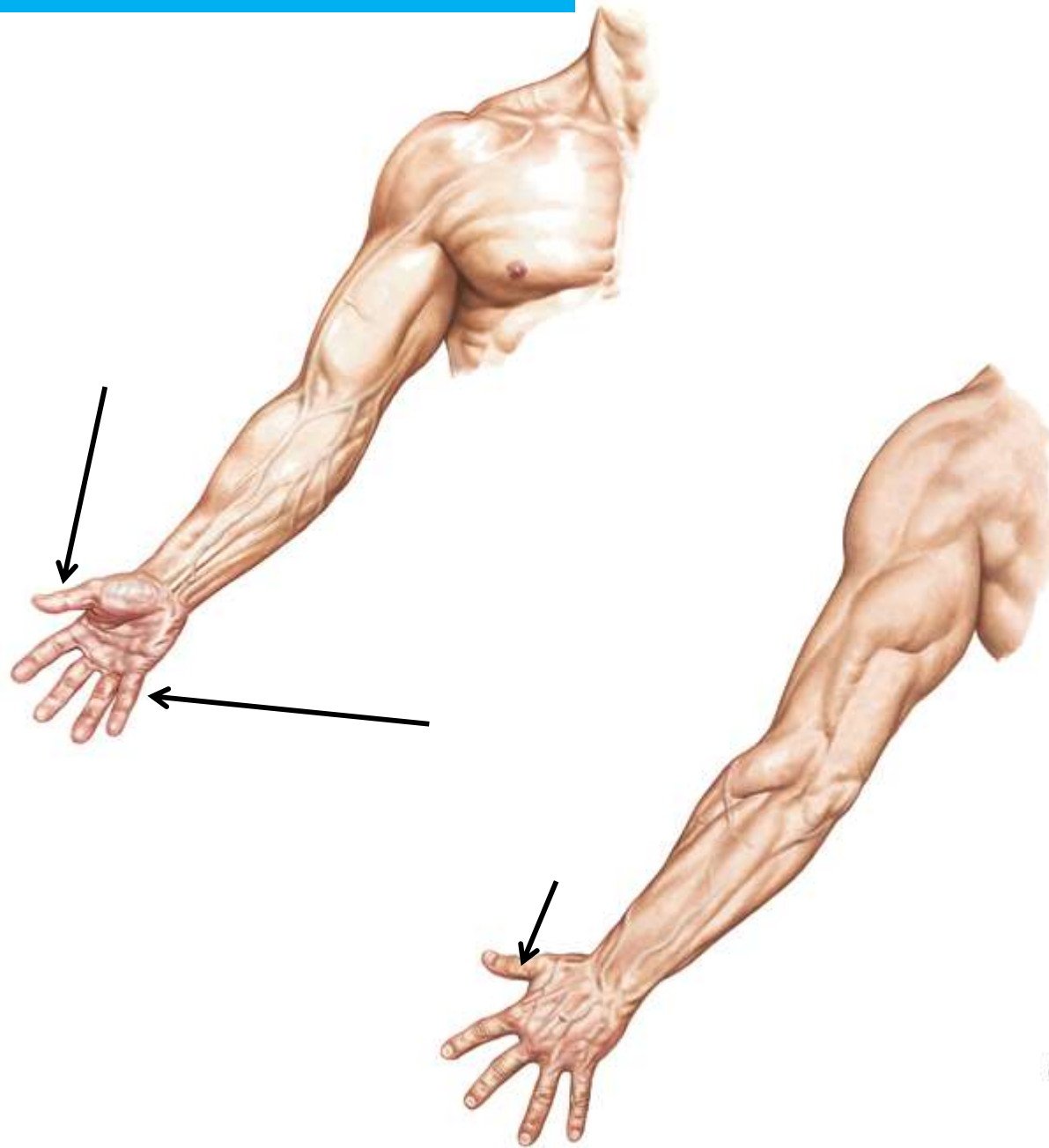
1. Which is medial kidney or aorta ?
2. What is superior to the kidney?



What is the relation of tendons to retinaculum

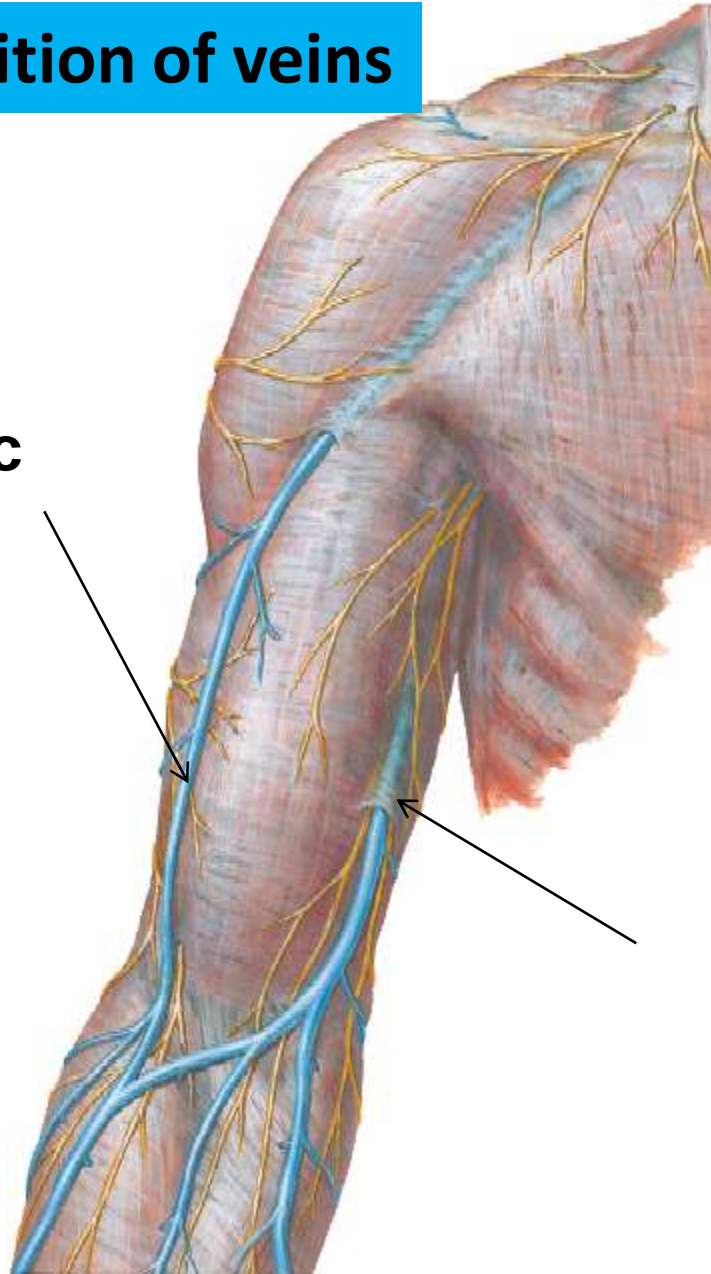


Which is medial thumb or little finger ?



Describe the position of veins

cephalic



basilic

Terms Used For Limbs

Terms used for limbs:

In the forearm:

radial or lateral & ulnar or medial.

In the hand:

palmar or anterior & dorsal or posterior.

In the leg:

fibular or lateral & tibial or medial.

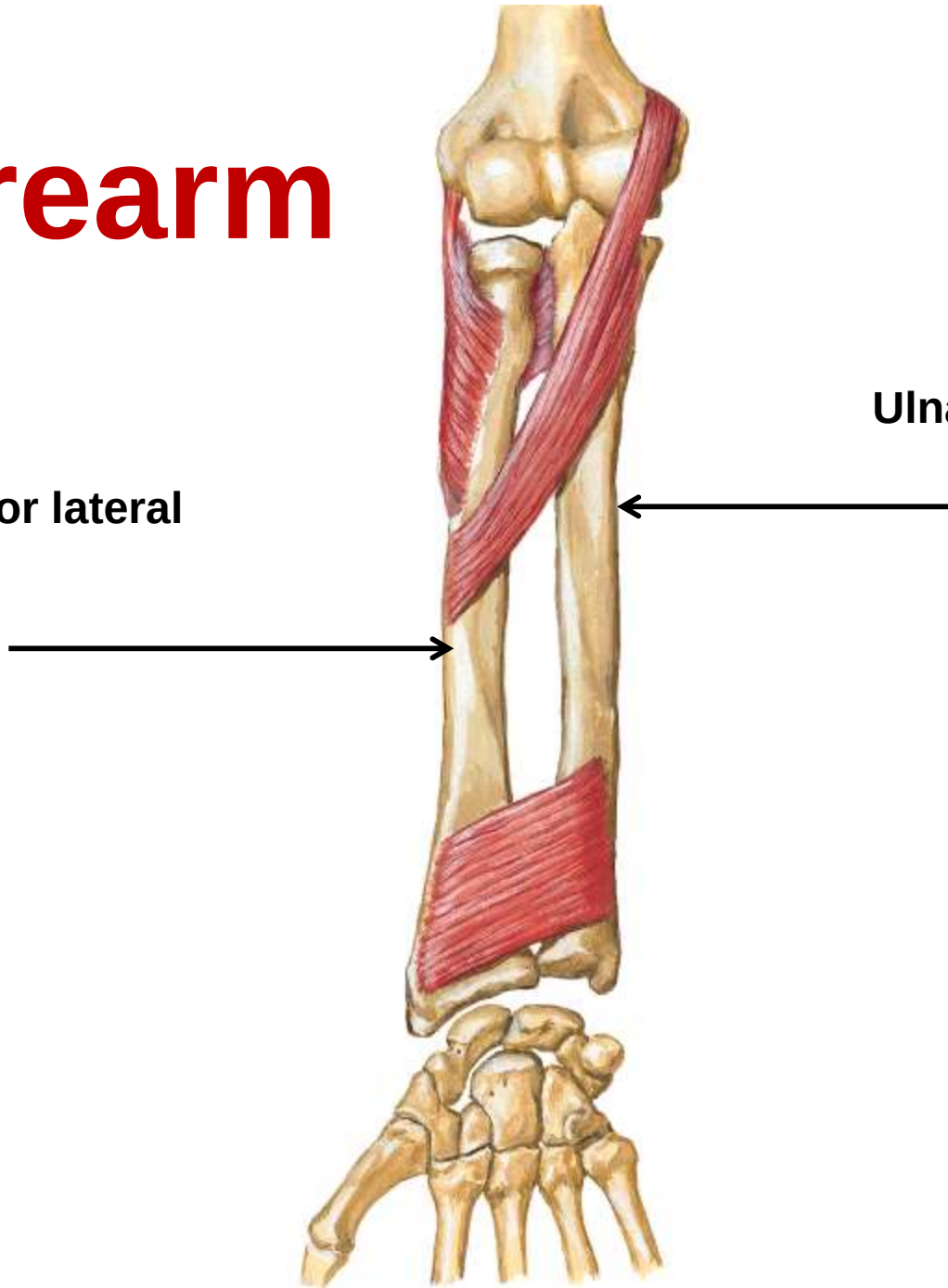
In the foot:

plantar or inferior & dorsal or superior.

Forearm

Radial or lateral

Ulnar or medial

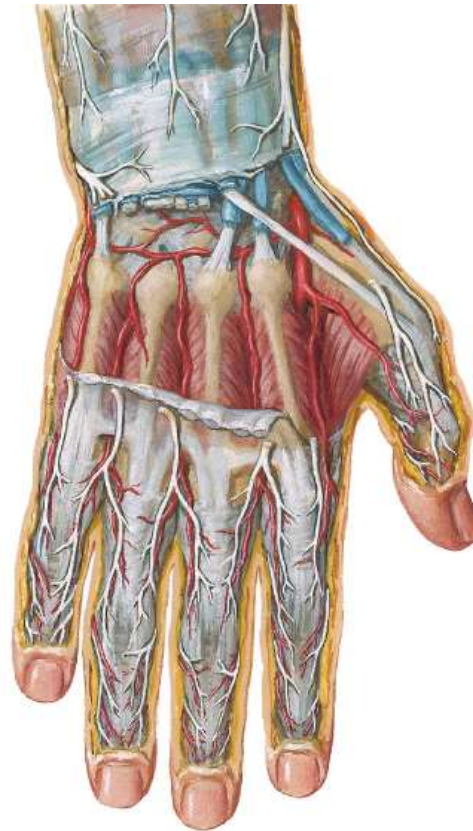


Hand

Palmar or anterior



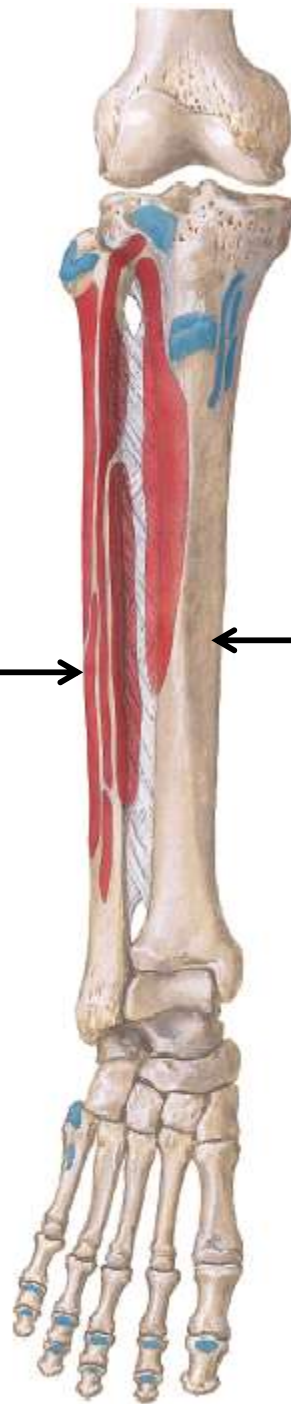
Dorsal or posterior



Leg

Fibular or lateral

Tibial or medial



Foot

Dorsum or superior



Planter or inferior



Terms of Movements

Flexion:

bending or approximation of two surfaces

Extension:

the return back of flexion

Abduction :

moving a limb away from the midline

Adduction:

moving a limb towards the midline

Adduction :of fingers

is approximating them together

Abduction: of fingers

is spreading them apart

TERMS OF MOTION



Flexion

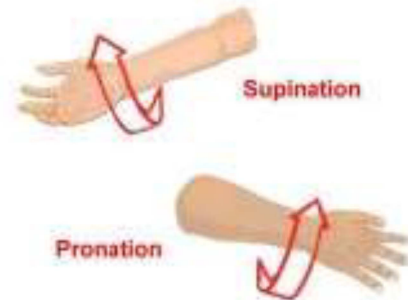


Extension



Medial Rotation

Lateral Rotation



Supination

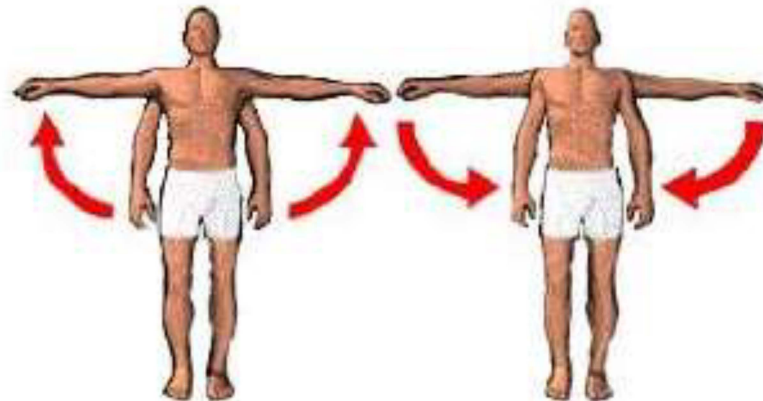
Pronation



Supine



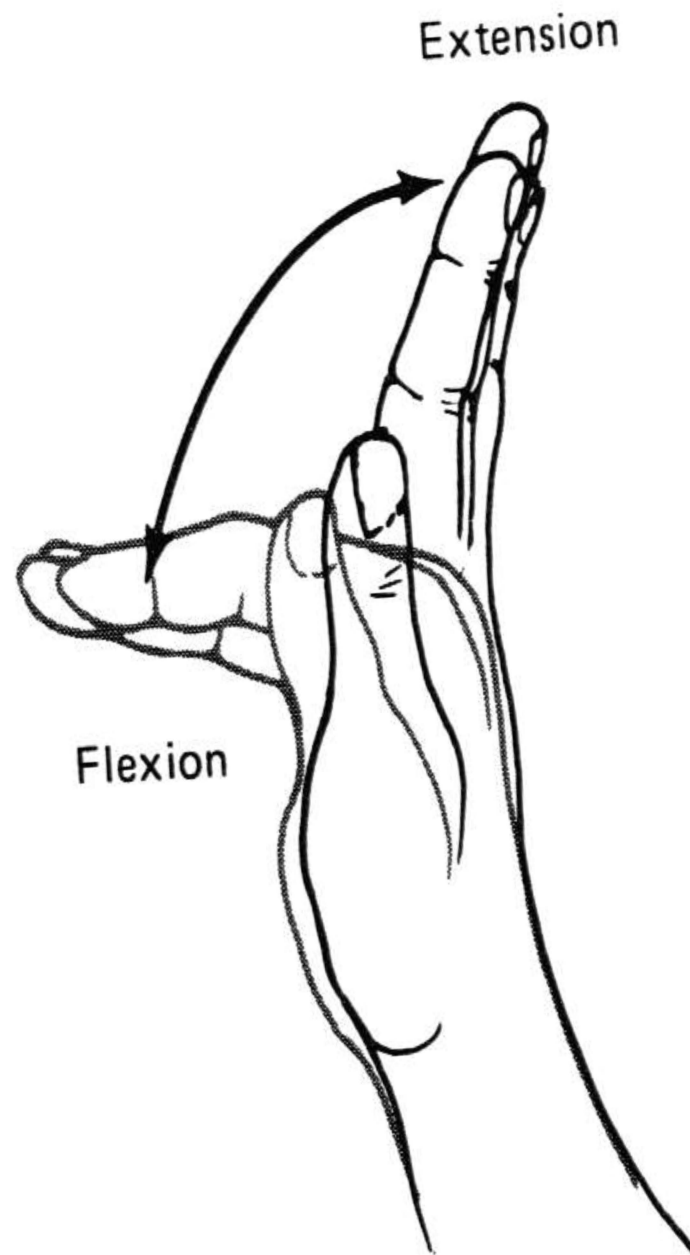
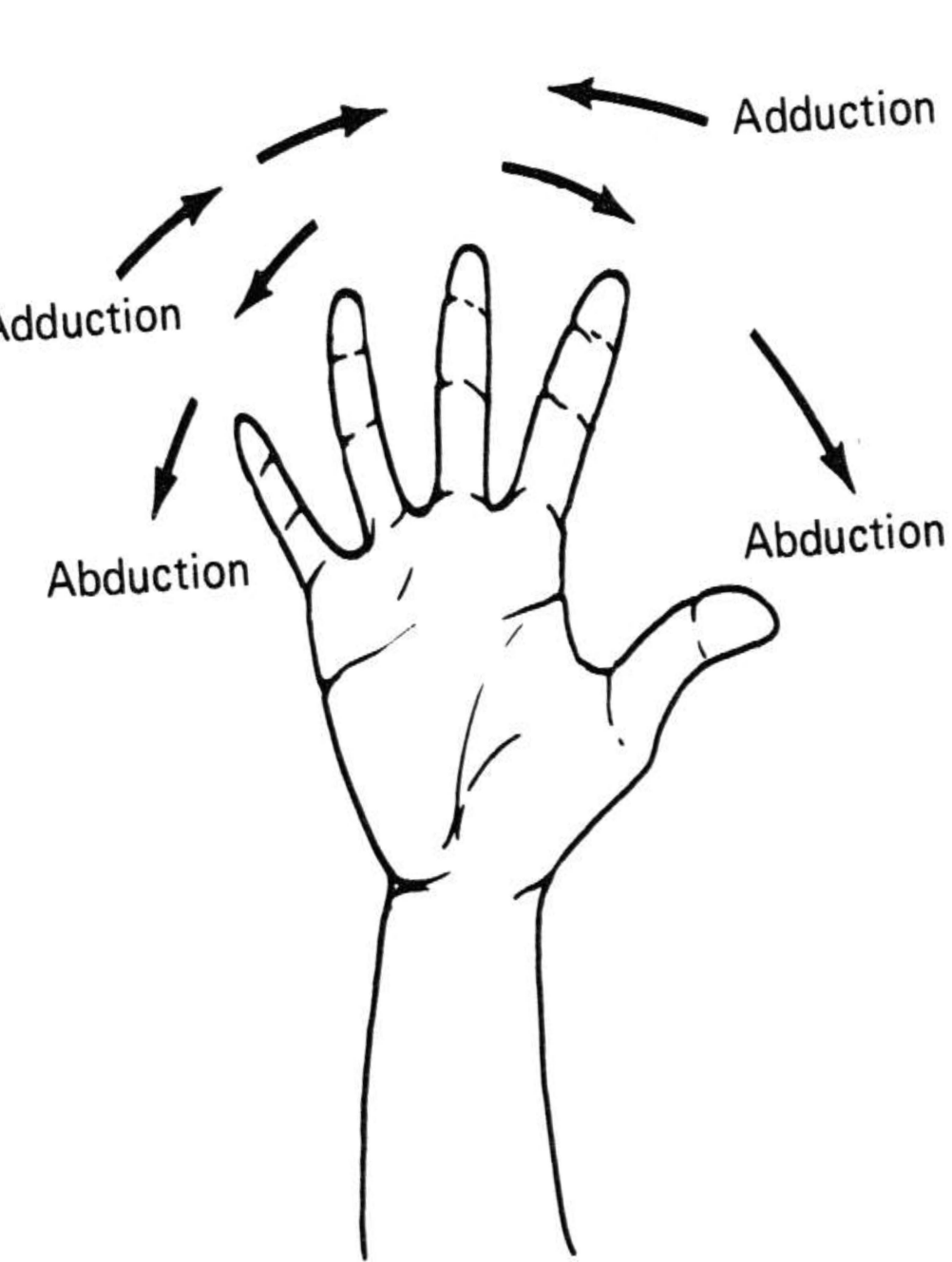
Prone



Abduction

Adduction





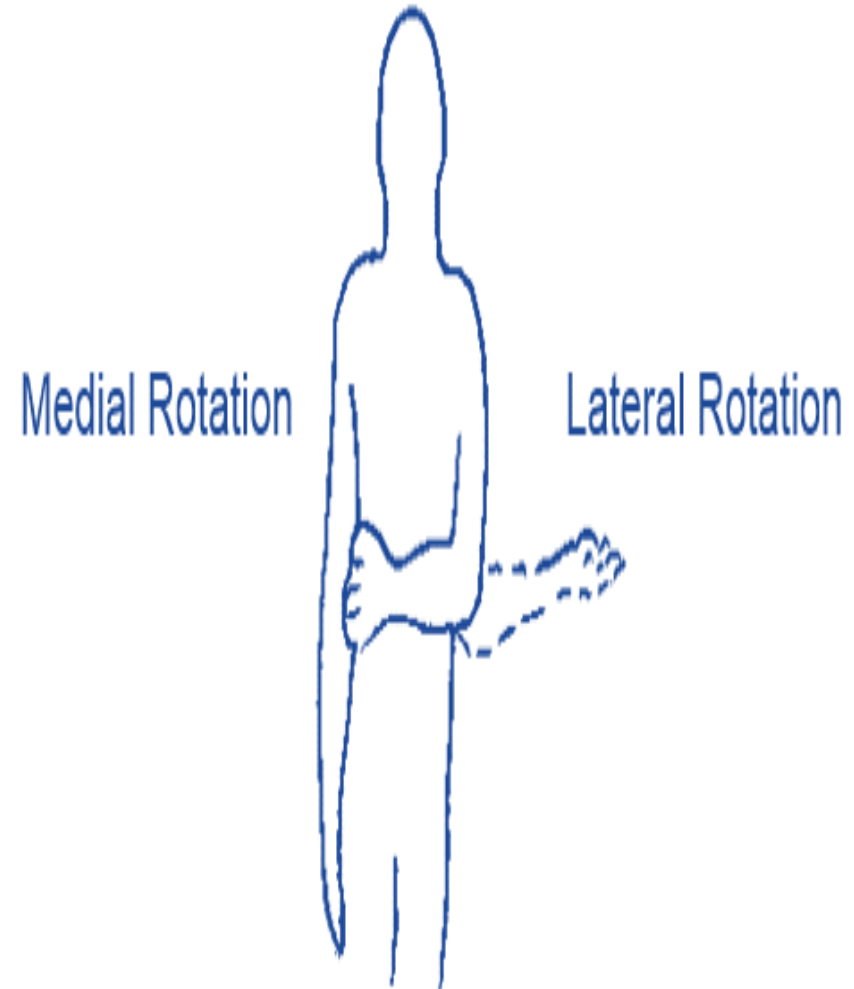
Terms of Movements

Medial rotation

turning the limb
along **its longitudinal**
axis **medially**

Lateral rotation

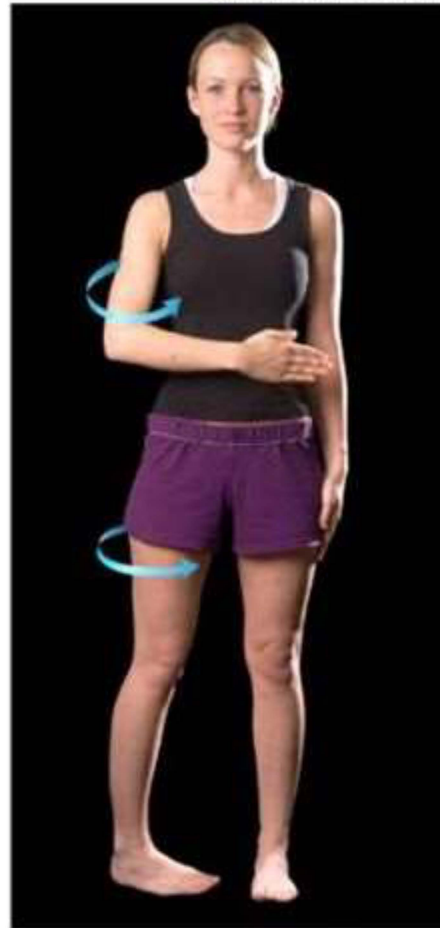
turning the limb
along its longitudinal
axis **laterally**



Rotation

- **Medial rotation**
- **Lateral rotation**

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(a) Medial (internal) rotation



(b) Lateral (external) rotation

Pronation

medial rotation of the forearm which brings the palm of hand to face backward

Supination

lateral rotation of forearm which brings the palm of hand to face forwards

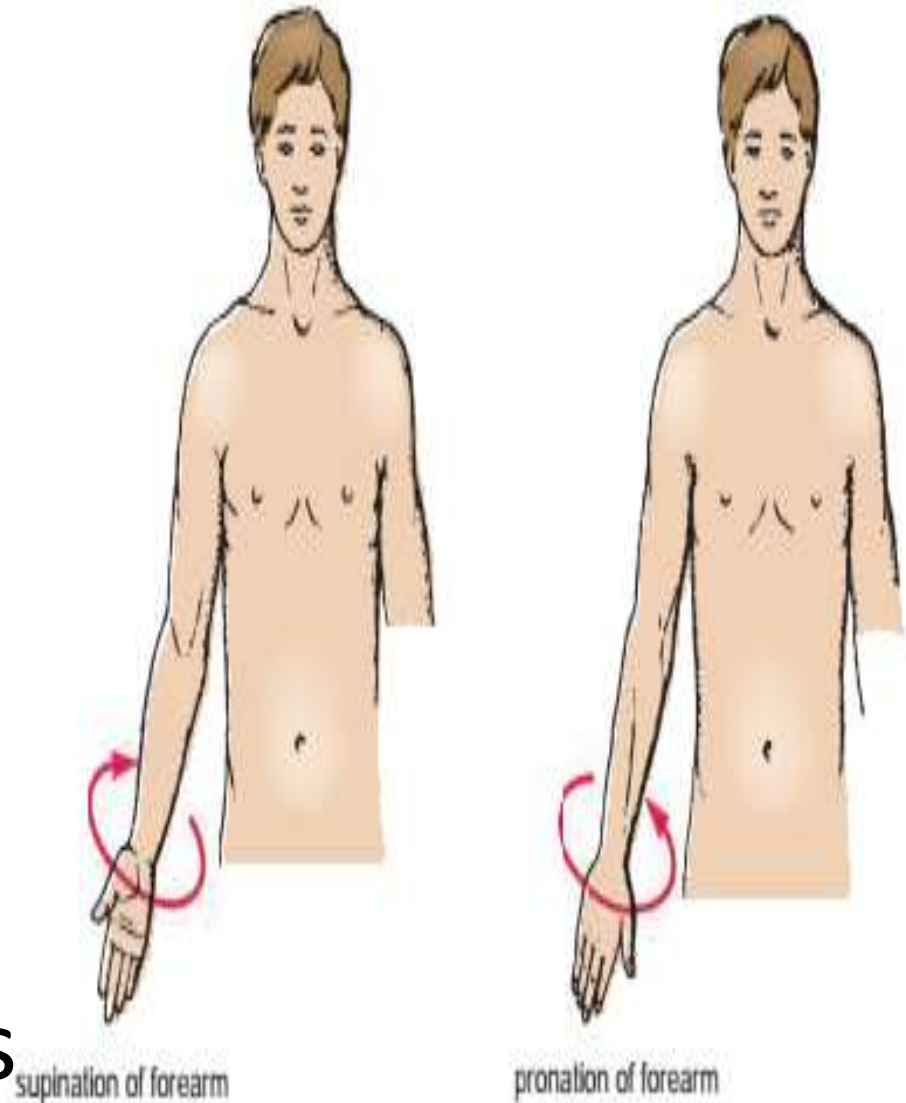


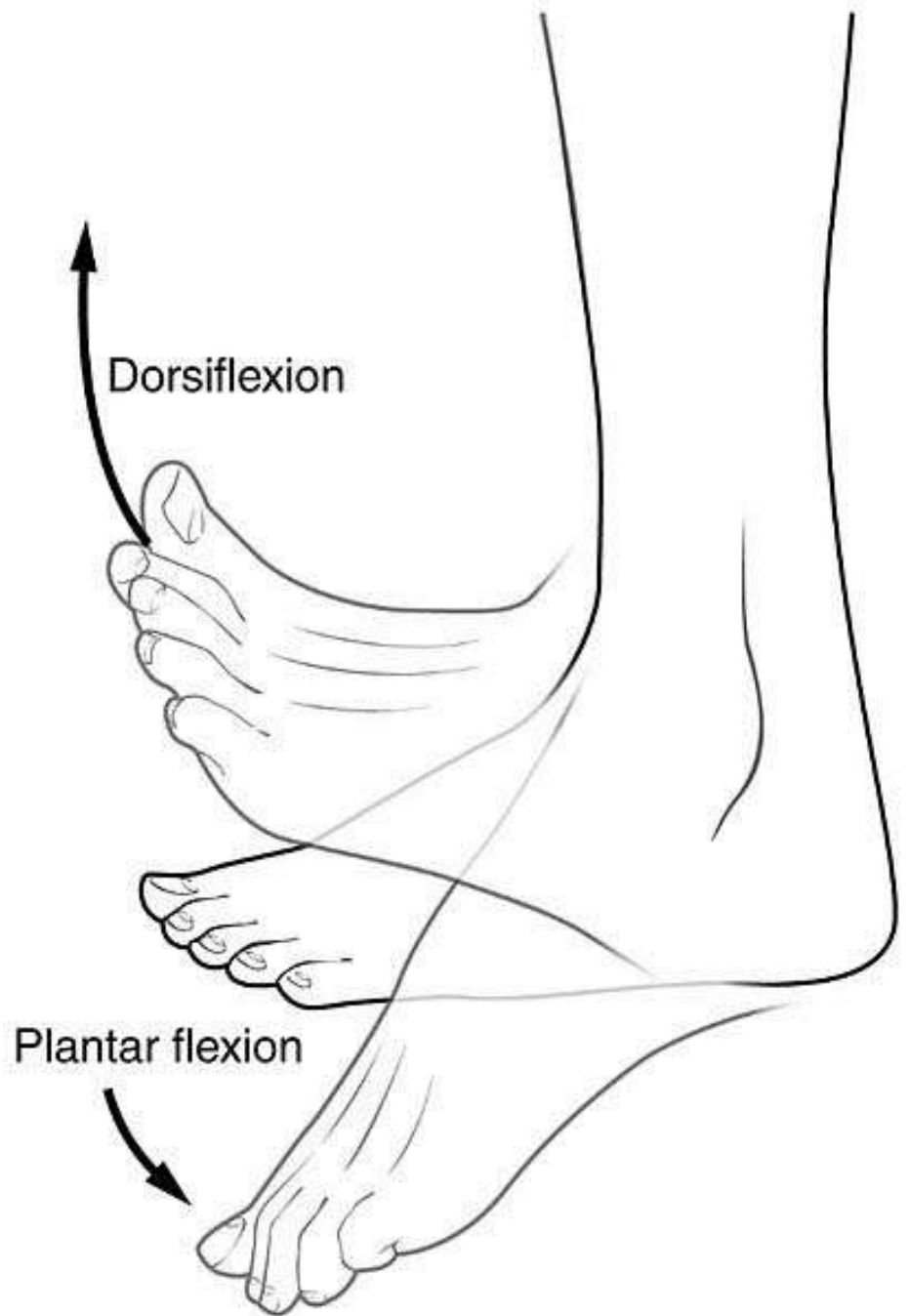
Figure 16

Dorsiflexion

bending the foot
at the ankle
upwards

Plantar flexion

bending the foot
at the ankle
downwards



Inversion

rotating the foot so
that the sole faces
inwards

Eversion

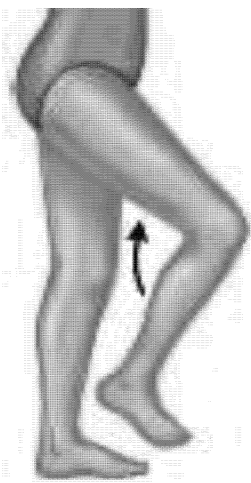
rotating the foot so
that the sole faces
outwards



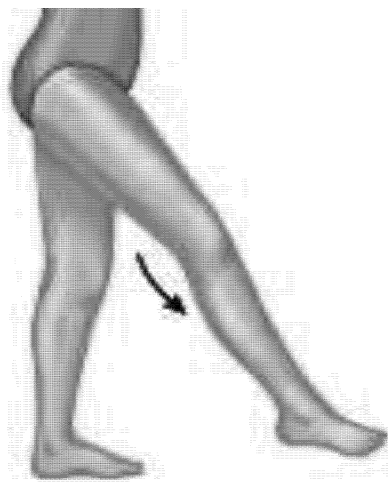
Special Movements

- Opposition
 - Touching the thumb to each finger.

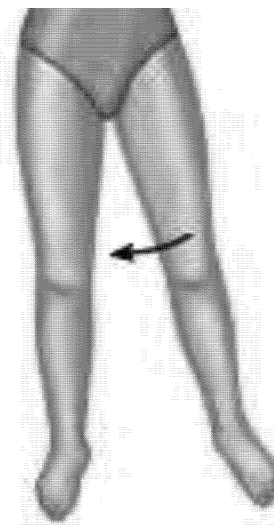




flexion of leg



extension of leg

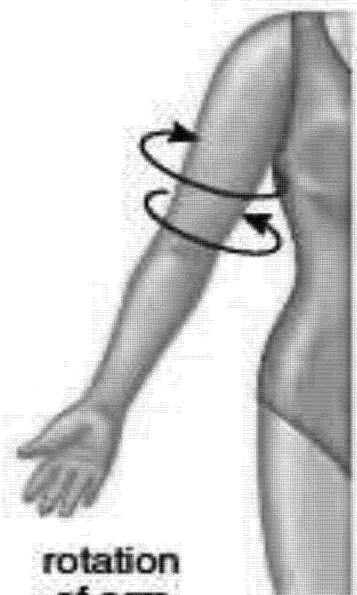


adduction of thigh

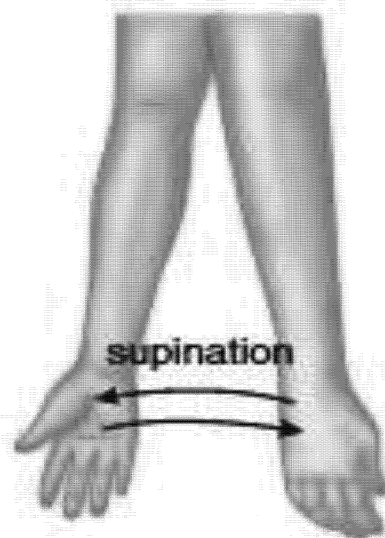


abduction of thigh

a. Angular movements



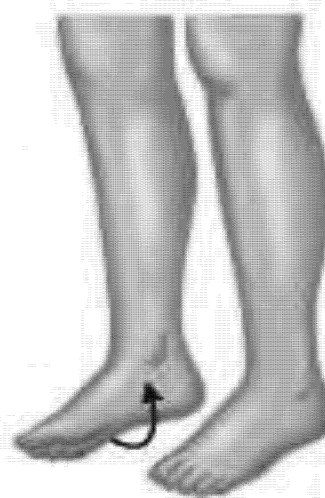
rotation
of arm



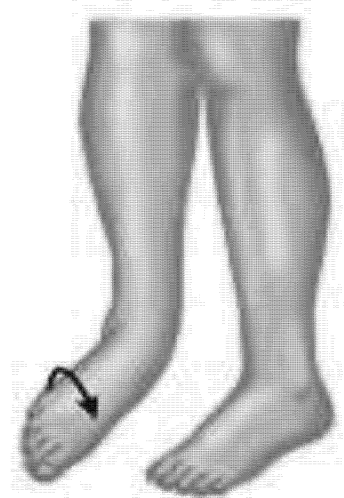
supination

pronation of hand

b. Circular movements



inversion



eversion

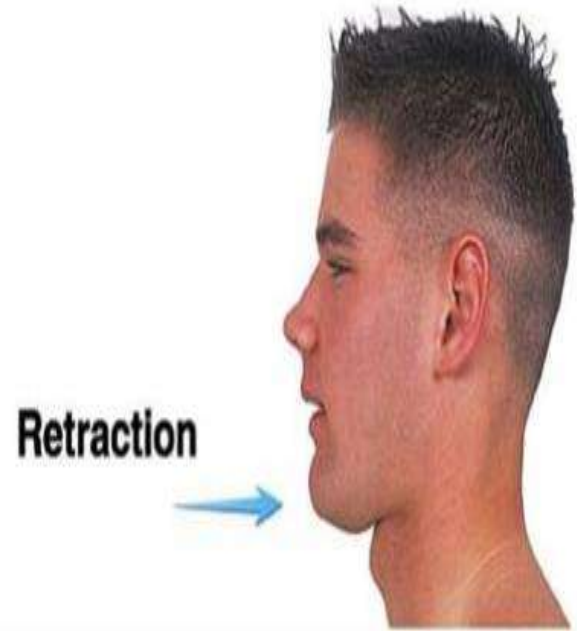
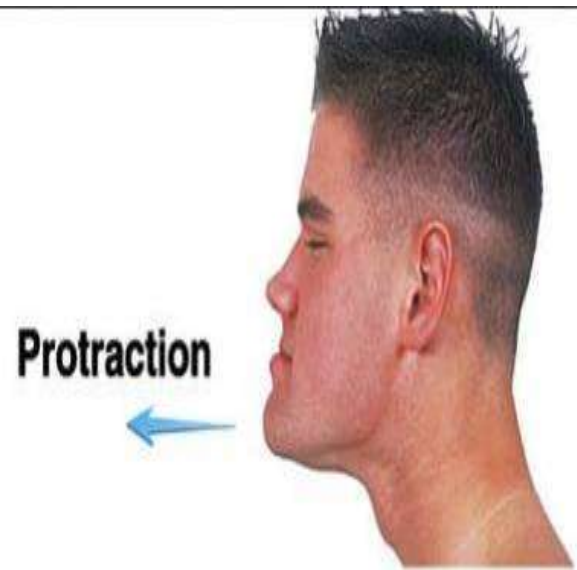
c. Special movements

Protraction

moving a
part forward
(protraction of
mandible or shoulder)

Retraction

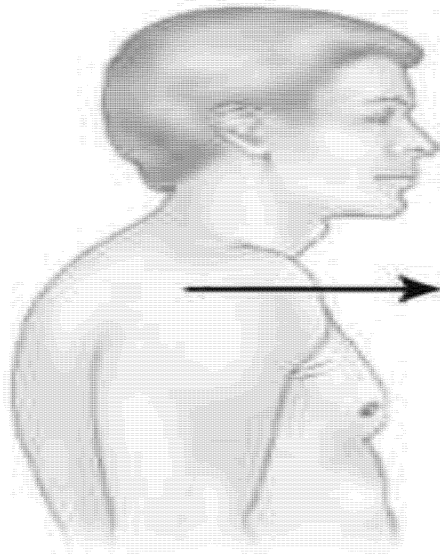
moving a part
backward
(retraction of
mandible or
shoulder)



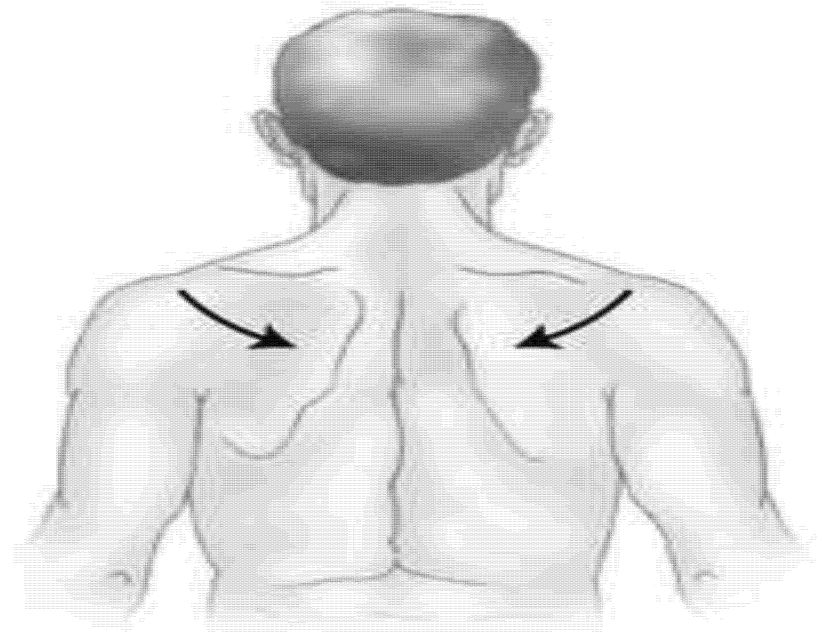
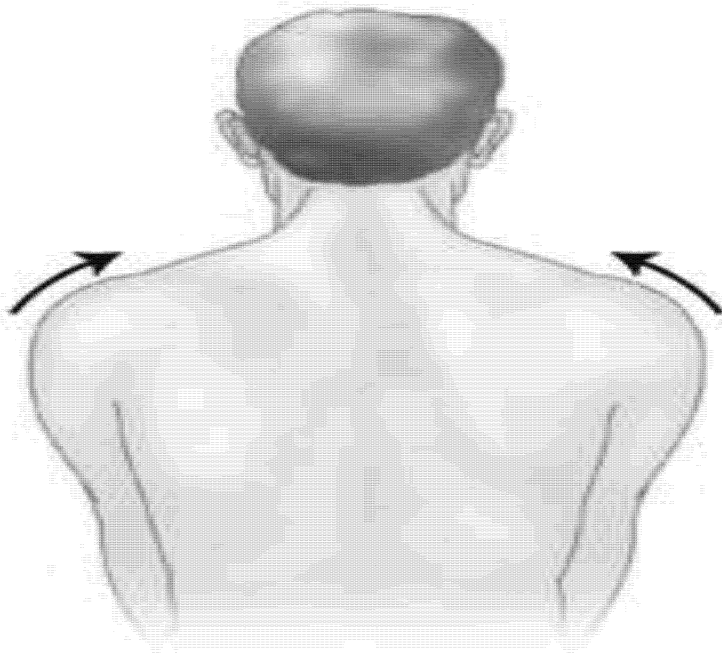
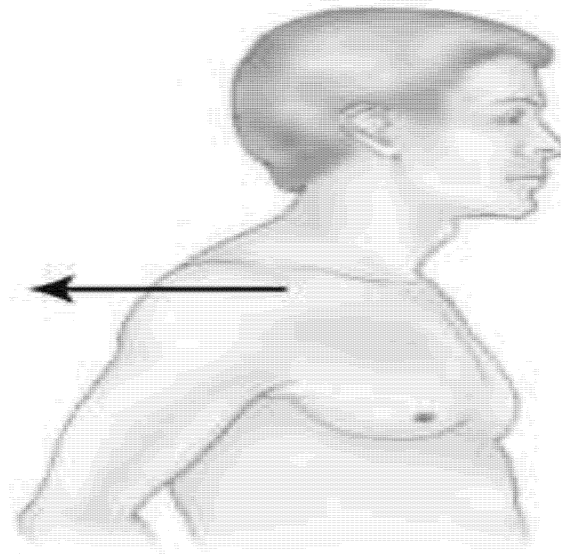
Shoulder Protraction



Protraction



Retraction



Elevation

moving a part upward
(elevation of shoulder or mandible)

Depression

moving a part downward
(depression of shoulder or mandible)

Elevation



Depression



